

```
import cv2
import matplotlib.pyplot as plt

# -----
# Stage 1: Read Image
# -----

image = cv2.imread("image.jpg")

# -----
# Stage 2: Preprocessing
# -----

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Remove noise using Gaussian Blur
blur = cv2.GaussianBlur(gray, (5, 5), 0)

# -----
# Stage 3: Edge Detection
# -----

edges = cv2.Canny(blur, 100, 200)

# -----
# Stage 4: Feature Extraction
# -----

# ORB feature detector
orb = cv2.ORB_create()

keypoints, descriptors = orb.detectAndCompute(
    gray, None
)

# Draw keypoints
feature_image = cv2.drawKeypoints(
```

```
        gray,
        keypoints,
        None,
        color=None,
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )

    # -----
    # Display all stages
    # -----

    plt.figure(figsize=(12, 10))

    plt.subplot(2, 2, 1)
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    plt.title("1. Original Image")
    plt.axis("off")

    plt.subplot(2, 2, 2)
    plt.imshow(blur, cmap="gray")
    plt.title("2. Preprocessed Image")
    plt.axis("off")

    plt.subplot(2, 2, 3)
    plt.imshow(edges, cmap="gray")
    plt.title("3. Edge Detection")
    plt.axis("off")

    plt.subplot(2, 2, 4)
    plt.imshow(cv2.cvtColor(feature_image, cv2.COLOR_BGR2RGB))
    plt.title("4. Feature Extraction")
    plt.axis("off")

    plt.tight_layout()
    plt.show()

    # Print feature information
    print("Number of Keypoints Detected:",
          len(keypoints))
```

```
if descriptors is not None:  
    print("Descriptor Shape:",  
          descriptors.shape)  
else:  
    print("No descriptors detected.")
```

1. Original Image



3. Edge Detection



2. Preprocessed Image



4. Feature Extraction



